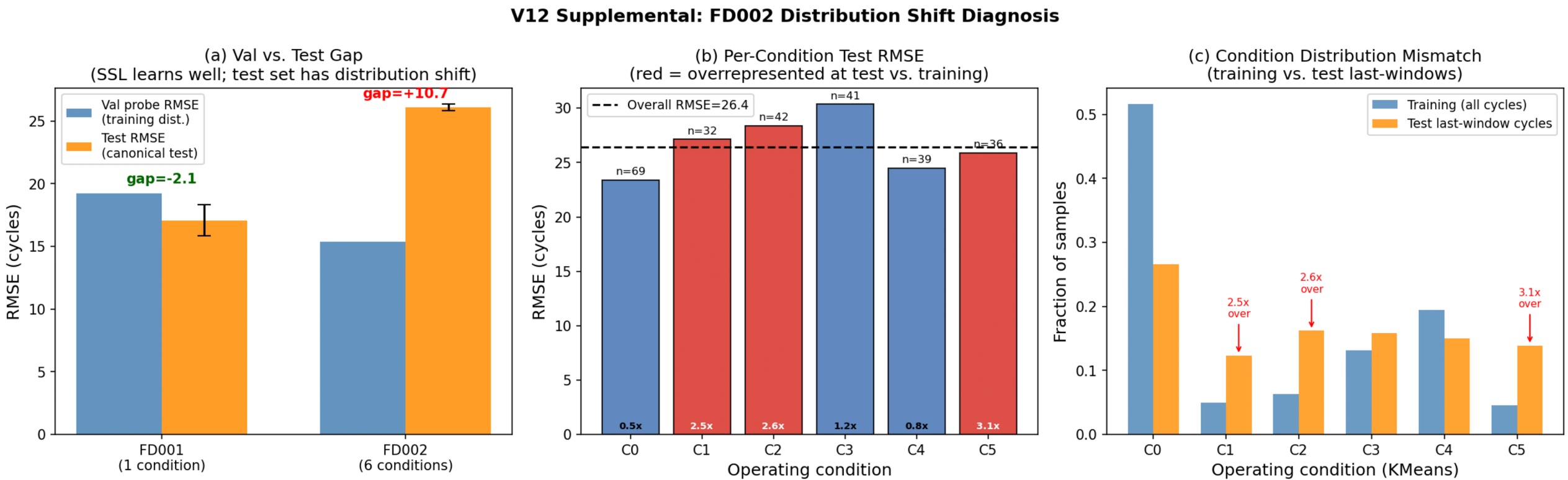
V12 Supplemental: FD002 Distribution Shift Diagnosis
(a) Val vs. Test Gap
(SSL learns well; test set has distribution shift)
Val probe RMSE (training dist.)
Test RMSE (canonical test)
gap=-2.1
gap=+10.7
RMSE (cycles)
FD001 (1 condition)
FD002 (6 conditions)
(b) Per-Condition Test RMSE
(red = overrepresented at test vs. training)
Overall RMSE=26.4
n=69
n=32
n=42
n=41
n=39
n=36
0.5x
2.5x
2.6x
1.2x
0.8x
3.1x
RMSE (cycles)
C0
C1
C2
C3
C4
C5
Operating condition
(c) Condition Distribution Mismatch
(training vs. test last-windows)
Training (all cycles)
Test last-window cycles
2.5x over
2.6x over
3.1x over
Fraction of samples
C0
C1
C2
C3
C4
C5
Operating condition (KMeans)